



भारतीय विज्ञान शिक्षा एवं अनुसंधान संस्थान मोहाली

(मानव संसाधन विकास मंत्रालय का एक स्वायत्त संस्थान, भारत सरकार के अधीन)

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MOHALI

(Estd. By Ministry of Human Resource Development, Govt. of India)

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NAME:

ROLL NUMBER:

LAB EXAM EES311

AIR WATER SOIL LABORATORY

Date: 13.11.2025

Total Marks: 30

Please read the questions carefully before answering and return the question paper with your name and roll number.

- 4) Explain what is meant by the accuracy error and the precision error (2 Marks).
- 2) For the meteorological practical you have measured $T=20^{\circ}\text{C}$ and $RH=90\%$. Calculate the saturation vapour pressure of water (p_{sat}) and actual vapour pressure $P = 0.61078 \exp\left(\frac{17.27T}{T + 237.3}\right)$ of water ($p_{\text{H}_2\text{O}}$) in kPa using the Thetens equation (1 Mark):
- 3) During the aerosol practical you learned about PM_{10} and $\text{PM}_{2.5}$. Which part of the instrument differs between a $\text{PM}_{2.5}$ and a PM_{10} analyzer/sampler? (1 Mark):
- 4) You are trying to determine the PM_{10} mass gravimetrically. Before you collected your sample you weighed the filter 10 times and got a weight of $0.1642 \pm 0.00003\text{g}$. After you retrieved the filter you re-weighed it for 10 times and got a weight of $0.1652 \pm 0.00004\text{g}$. You collected $5 \pm 0.001 \text{ m}^3$ of air during the sampling. Calculate the PM_{10} mass loading and their uncertainty in the units $\mu\text{g}/\text{m}^3$ (2 Marks).
- 5) Name the GC columns used for analyses of C2-C5 and C6-C10 hydrocarbons respectively in the TD-GC-FID method that was part of your laboratory practical ((2 Marks).
- 6) If the 8 h average measured mixing ratio of carbon monoxide (CO) is 1000 ppb at 10 degree Celsius and 1 atm pressure, determine whether the value exceeds the national ambient air quality of 2mg m^{-3} for CO. (2 Marks)
- 7) You have estimated parameters such as turbidity, ~~sulphate~~ ^{Phosphate}, total dissolved solids, COD, nitrate and hardness. Mention which principle (e.g. gravimetric; light-scattering; light-absorbing; acid-base, oxidation-reduction, complexometric titration etc.) you have used for estimation of each of these parameters. If a parameter has been estimated using more than one method, separately mention that method. (1.5 Marks)
- 8) What is difference between Electrical conductivity (EC) and total dissolved solids (TDS)? Cite a major limitation each of the EC and TDS analysis. (1 +1 Marks)
- 9) Can turbidity provide a good estimate of total suspended solids? Why or why not? (1+ 1 Marks)
- 10) Which five water quality parameters would you test for water supply and which five would you examine for wastewater monitoring, given the IISER Mohali campus's water supply and wastewater generation? Give justification for your chosen parameters. (2.5 Marks)
- 11) What are the different reagents used in COD determination? Write function of each chemical used? What is the difference between closed reflux and open reflux method of COD determination? (1.5 + 0.5 Marks)
- 12) A food safety inspector visits a bakery to check for possible contamination. Instead of selecting samples randomly, the inspector chooses pastries that look stale, have unusual colouring, or are stored near cleaning chemicals.
 - a) What sampling method is the inspector using?

- b) Why might the inspector choose this approach?
- c) Discuss the potential benefits and drawbacks of relying on this method for assessing the overall safety of the bakery's products. (1 mark)
- 13) During your environmental chemistry lab, you used an ICP-MS instrument to analyze trace metals in groundwater samples. The instrument has a detection limit of $0.01 \mu\text{g/L}$ for lead (Pb). You measured lead concentrations in one sample and obtained the following readings: $0.012 \mu\text{g/L}$, $0.011 \mu\text{g/L}$, $0.013 \mu\text{g/L}$.
- a) Based on these results, was the ICP-MS sensitive enough to detect lead in this sample?
- b) If another sample had a lead concentration of $0.008 \mu\text{g/L}$, would the same ICP-MS setup be able to detect it reliably? (1 + 1 mark)
- 14) What is the principle of heavy liquid separation, and how do variations in mineral composition provide information on provenance interpretation? (2 marks)
- 15) How are organic compounds identified and quantified using GC-MS? (2 marks)
- 16) What does the grain size distribution indicate about sediment transport and depositional energy conditions? (2 marks)
- 17) Using the cumulative grain-size distribution curve, where Φ_5 , Φ_{16} , Φ_{84} , and Φ_{95} are -1 , 0 , 2 , and 3 respectively, calculate the sorting (σ_1) of the sediment and classify it according to Folk and Ward (1957) (1 mark)